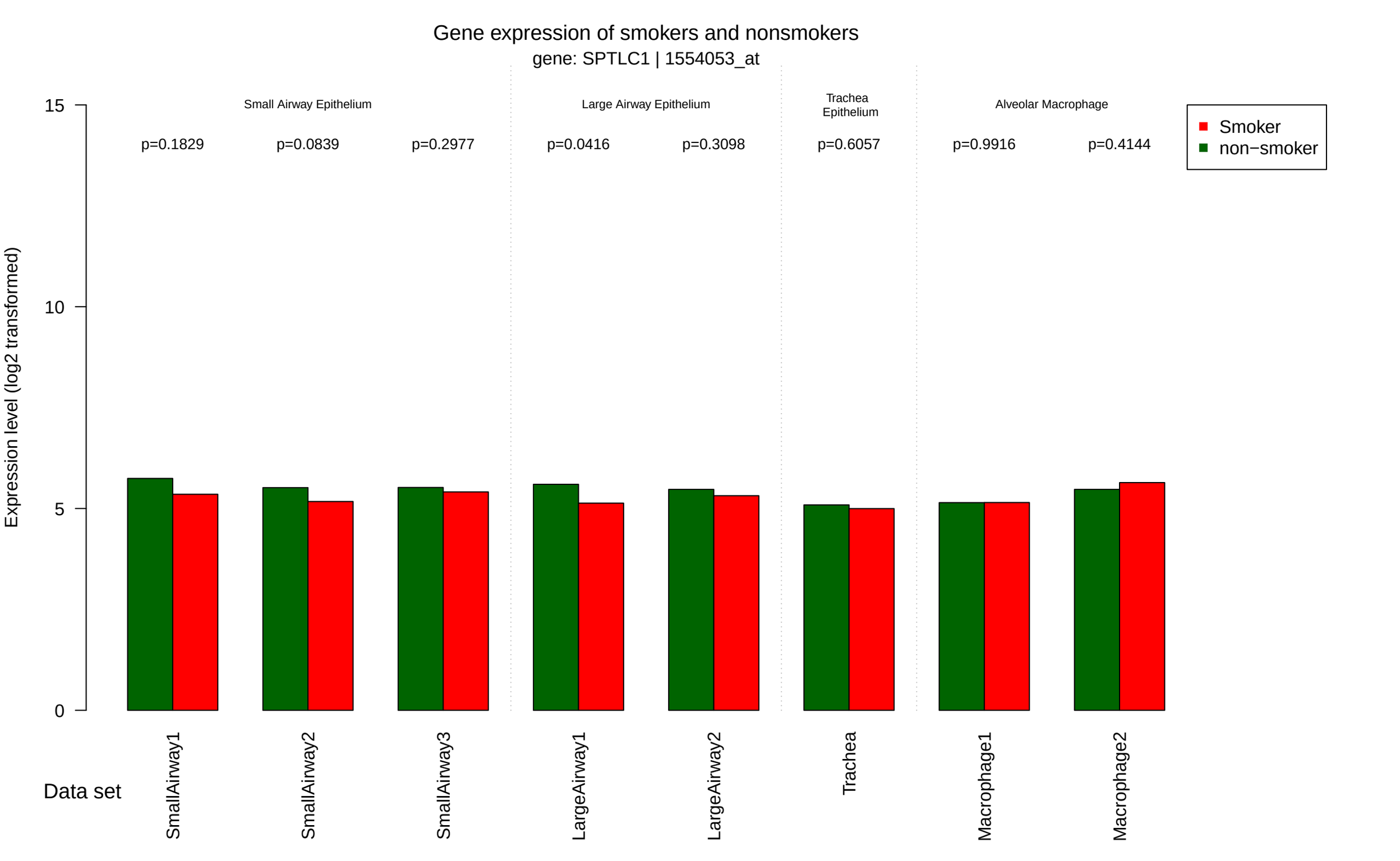
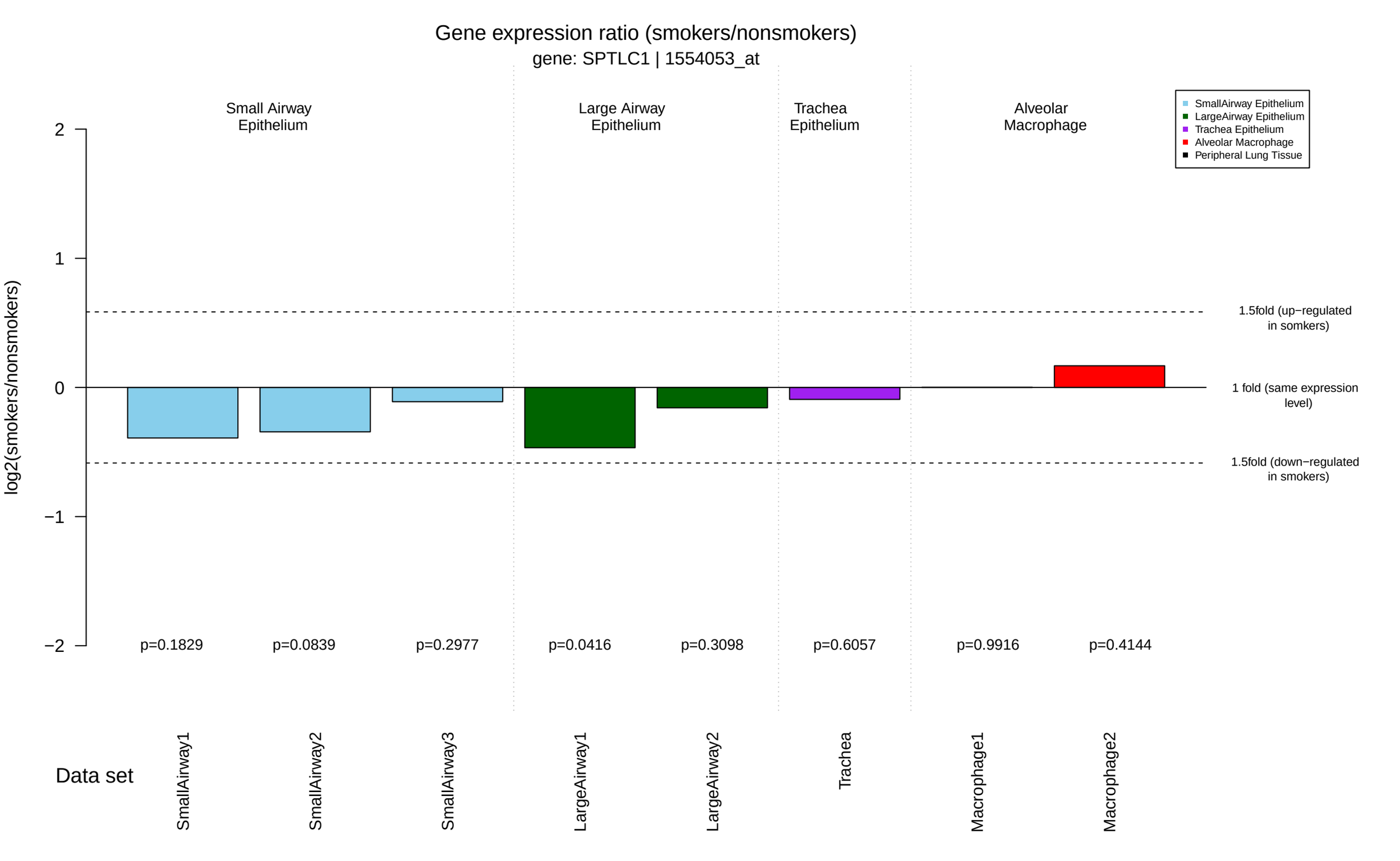
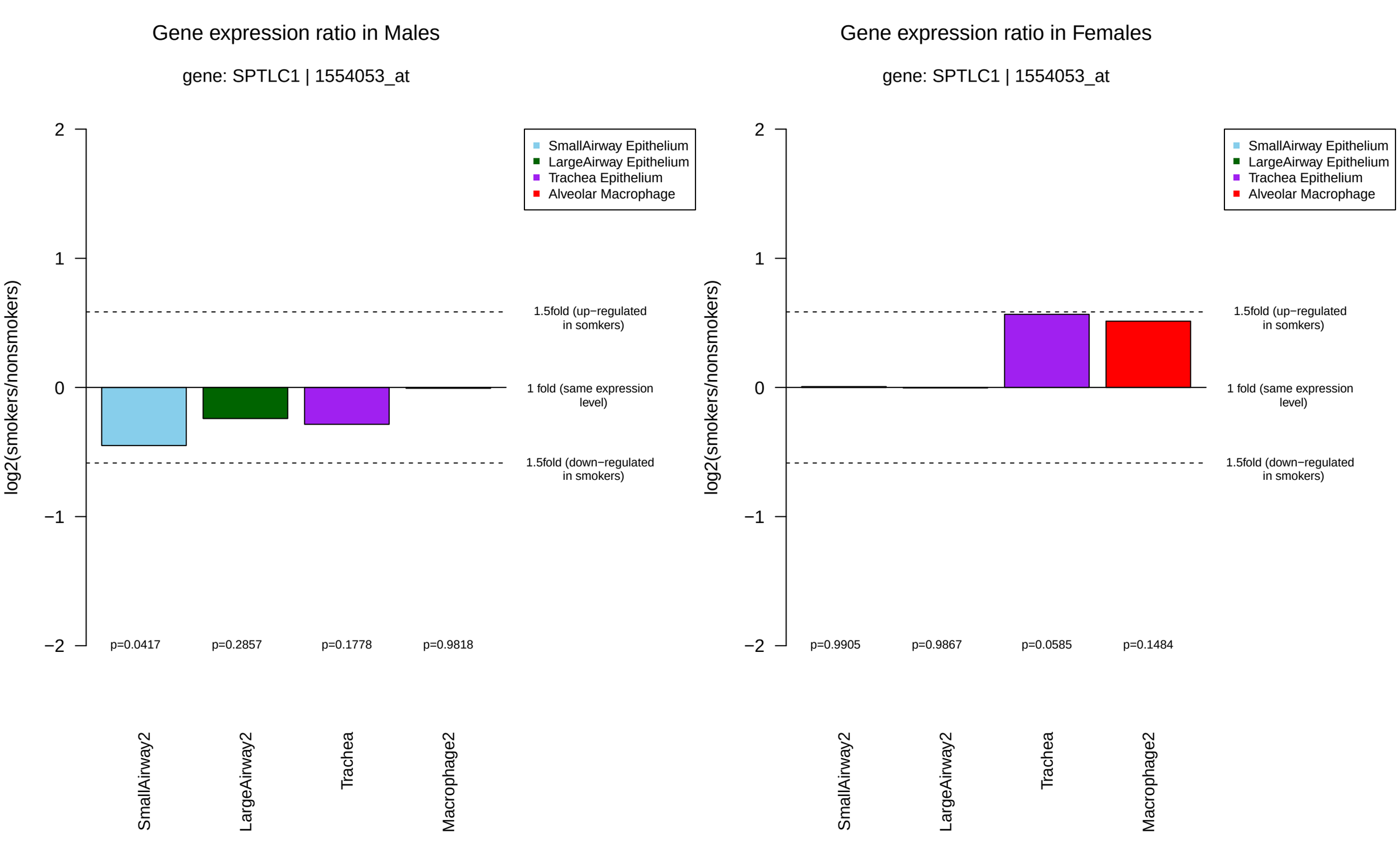
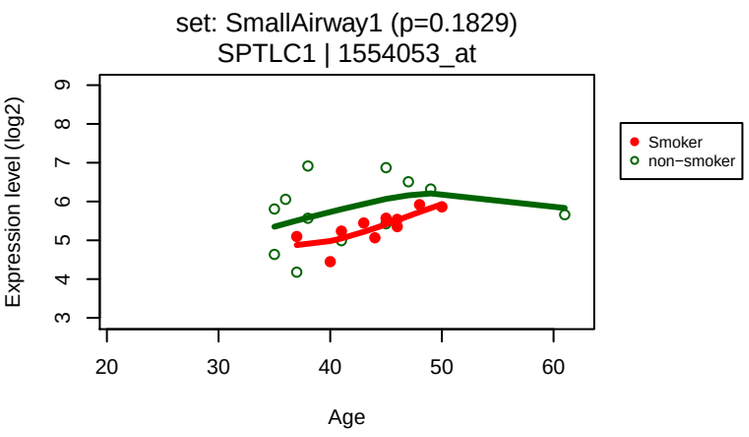
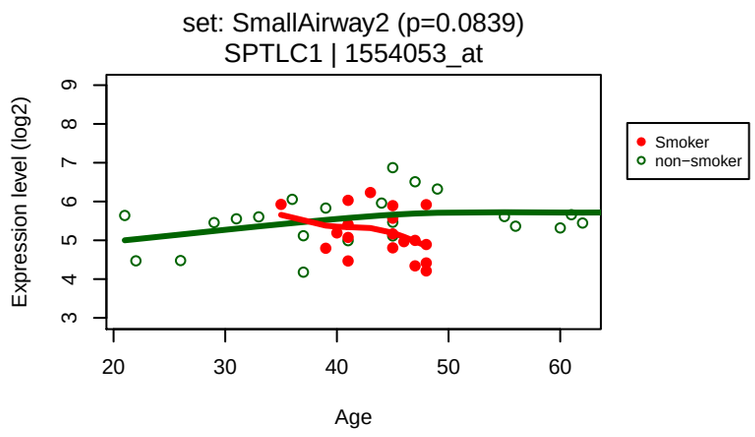
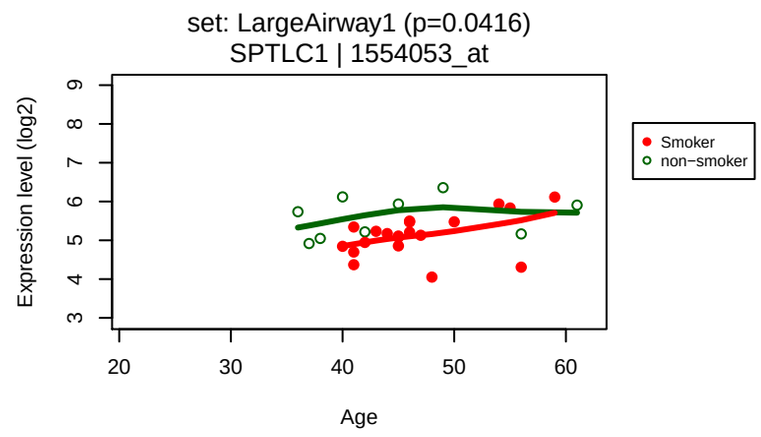

Gene expression of smokers and nonsmokers
gene: SPTLC1 | 1554053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1829
p=0.0839
p=0.2977
p=0.0416
p=0.3098
p=0.6057
p=0.9916
p=0.4144
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

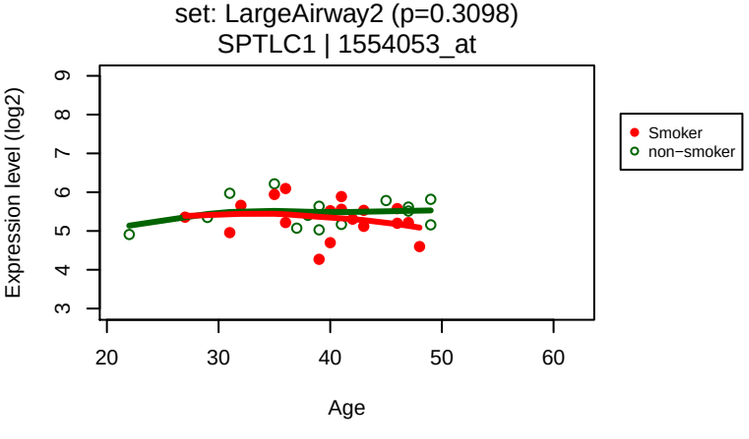

Gene expression ratio (smokers/nonsmokers)
gene: SPTLC1 | 1554053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1829
p=0.0839
p=0.2977
p=0.0416
p=0.3098
p=0.6057
p=0.9916
p=0.4144
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

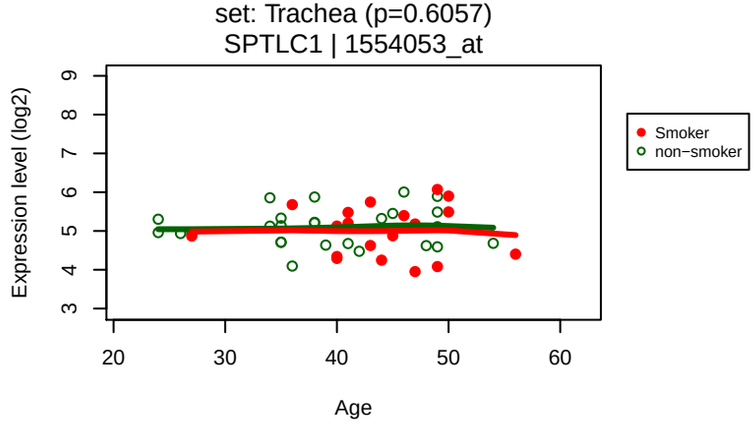

Gene expression ratio in Males
gene: SPTLC1 | 1554053_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.2857
p=0.1778
p=0.9818
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTLC1 | 1554053_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9905
p=0.9867
p=0.0585
p=0.1484
SmallAirway2
LargeAirway2
Trachea
Macrophage2

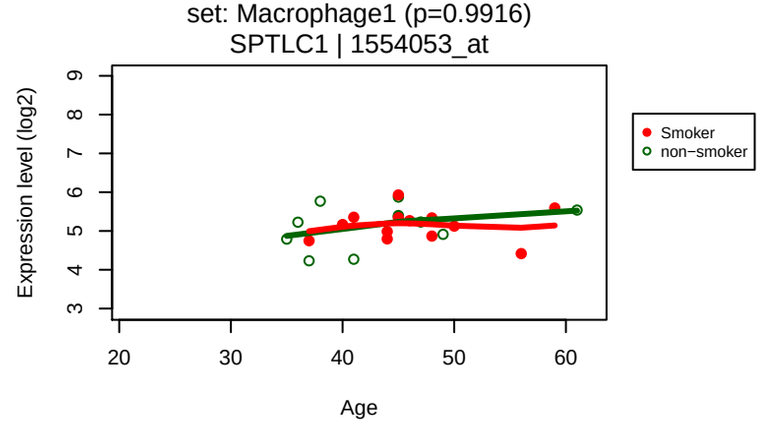

set: SmallAirway1 (p=0.1829)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker

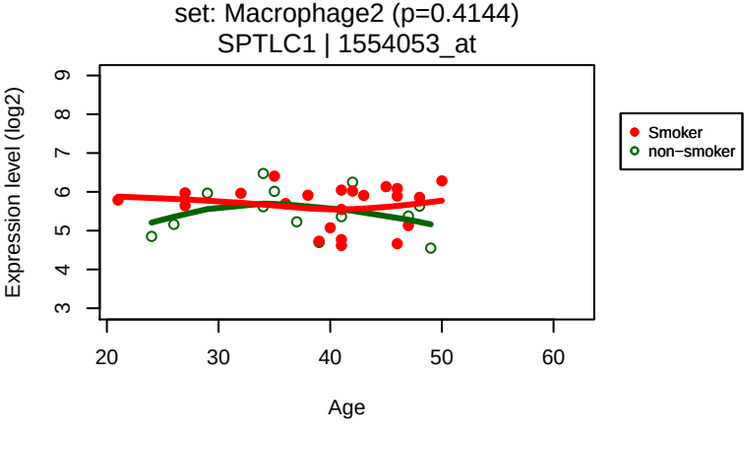

set: SmallAirway2 (p=0.0839)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker

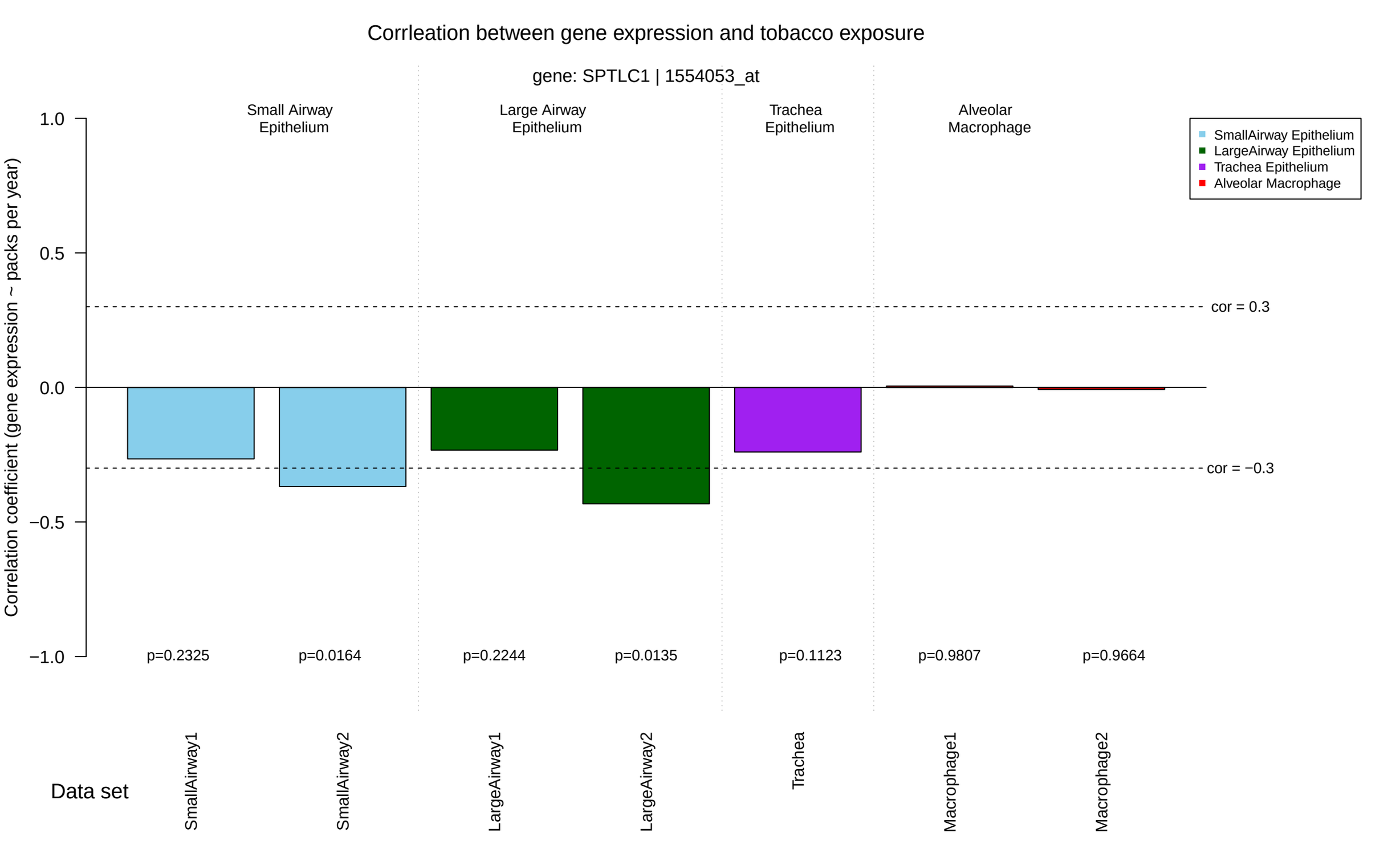

set: LargeAirway1 (p=0.0416)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3098)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6057)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9916)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4144)
SPTLC1 | 1554053_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTLC1 | 1554053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2325
p=0.0164
p=0.2244
p=0.0135
p=0.1123
p=0.9807
p=0.9664
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2